	Biomedical Limited, No 1 Ohimege Road, Industrial Estate, Ilorin Kwara State, PMB 1449			MASTER COPY STAMP HERE
	Title: **Standard Operating Procedure for Preventive Maintenance of Utility and Production Equipment**			
Department	**Unit**	**BML-ENG-NG-003**	**Revision No.:**	06
			Effective Date:	**April, 2025**
ENGINEERING	ENGINEERING GENERAL		**Next Review Date:**	**March, 2027**
Prepared By:	**Checked By:**		**Approved By:**	
sign/date	**sign/date**		**sign/date**	

1.0 OBJECTIVE:

- To lay down the procedure for Preventive Maintenance of Utility and Production Equipment in Biomedical Limited

2.0 SCOPE:

- This SOP shall be applicable for preventive maintenance of all types of equipment and systems in Biomedical Limited

3.0 RESPONSIBILITIES/ACCOUNTABILITIES:

- It is the responsibility of any qualified personnel
- It is the responsibility of Biomedical Limited to provide PPE for the qualified employee performing the task
- It is the responsibility of contactors to provide PPE for their qualified staff
- It is the responsibility of the Engineering head to ensure that the SOP is strictly followed and that the right PPE are used

4.0 ABBREVIATION AND DEFINITIONS:

- SOP: Standard Operating Procedure.
- PPE: Personal Protective Equipment.
- NA: Not Applicable.
- GMP: Good Manufacturing Practices.

5.0 PROCEDURE:

- Preventive maintenance plan shall be prepared and deployed to all employees.
- The new equipment installed shall be incorporated in the preventive maintenance plan.
- All scheduled preventive maintenance shall be carried out as per the scheduled date.
- Preventive maintenance schedule shall be carried out as per the scheduled date.
- Preventive maintenance checklist BML/ENG/NG/G0001 shall be prepared based on past experience, maintenance manuals and manufacturers recommendation.
- The Preventive maintenance checklist) number is an alphanumeric system comprising of 20 characters.
- Ensure the right quality of consumables including oil & grease is available including the needed spare parts.

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- Ensure that the main switch of the machine is turned OFF, the fuses have been removed wherever available and utilities are isolated whenever required. Tag UNDER MAINTENANCE label to the equipment under maintenance also perform lock out tag where applicable.
- Preventive maintenance should be carried out as per the preventive maintenance checklist with form number: BML/ENG/NG/003/G0001 and tick-mark the checks. In case a defect is noticed during inspection, document such defect on the space provided on the preventive maintenance equipment checklist, fix immediately and in a case whereby parts are not available or personnel did not have the capabilities to fix the defects. Such Defect should be documented in open defect space on the equipment checklist (in annexure 1) and maintenance personnel should also communicate to the Senior Maintenance Engineer, the Senior Maintenance officer or the Engineering Head for proper auctioning, always monitor the progress of your report.
- If the defects were not captured during the cause of performing the preventive maintenance, such defects should be recorded inside the defect handling sheet BML-ENG-NG-003-F05 closer to the machine
- In case if repeated breakdowns observed during operation, corrective actions should be planned during preventive maintenance meeting.
- After completing the preventive maintenance, take machine trial and remove the UNDER-MAINTENANCE tag.
- If the machine status is satisfactory, handover the machine to the respective department representative in case of utility and HVAC equipment, performance to be verified by responsible engineer.
- Symbolically update preventive maintenance schedule and plan as indicated below
- Frequency= Weekly (W), Monthly (M), Quarterly (Q), Annually (A)

  Executed and okay=☑
  Executed with defect=☒
  Not done= ☐ (blank)
  Rescheduled= R
- In case of any major part to be replaced, which may affect the machine performance should take place through a change control procedure.
- In case the machine is not available for preventive maintenance due to production plan, the preventive maintenance can be rescheduled within 7 days of the scheduled date.
- If the machine is not put into use or not used in the complete month, the preventive maintenance shall be carried out once in three months to keep the machine in healthy condition.

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- As a maintenance personnel or an equipment owner, always refer to the equipment handover booklet/usage booklet (infusion engineering handover booklet, water treatment handover booklet, autoclave operator handover booklet and boiler operator handover booklet with document number: BML-ENG-NG-003-F01, BML-ENG-NG-003-F02, BML-ENG-NG-003-F03, BML-ENG-NG-003-F04 respectively) as this will give you more insights about the equipment.

6.0 REFERENCES AND ENCLOSURE:

- Equipment Operation Manual.
- Equipment Preventive maintenance checklist -BML/ENG/NG/003/G01
- Defect Handling Sheet -BML-ENG-NG-003-F05
- Infusion engineering department handover booklet -BML-ENG-NG-003-F01
- Water treatment handover booklet -BML-ENG-NG-003-F02
- Autoclave operator handover booklet -BML-ENG-NG-003-F03
- Boiler operator handover booklet -BML-ENG-NG-003-F04

7.0 TRAINING INVOLVEMENT:

All Engineering staff
All machine Operators

8.0 DISTRIBUTION AND COPY CONTROL:

- This is the various department/section where the SOP copies will be utilized.

MASTER COPY	QUALITY
CONTROLLED COPY	QUALITY, ENGINEERING, PRODUCTION

9.0 CHANGE HISTORY

CURRENT REVISION	SUPERSEDE	REASON FOR REVIEW
00	Nil	First issue
01	00	To review
02	01	To review
03	02	To update as per NAFDAC Observation
04	03	To update
05	04	To update

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06	05	• Change in sop format. • Inclusion of traceability of enclosures to parent SOP

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